

PDA (Push Down Automata)

↳ NFA with stack

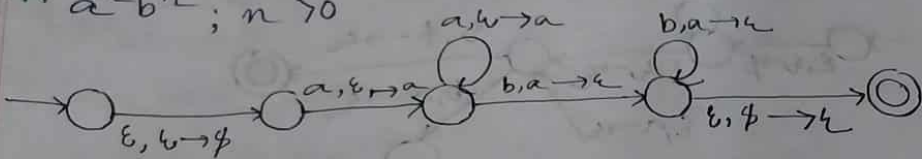
A, B

Pop this
Symbol

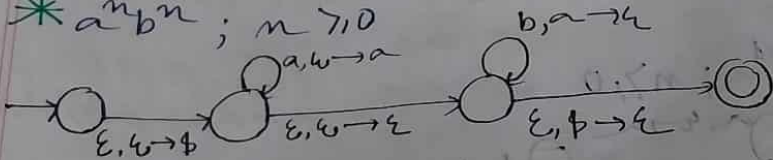
→

↓
Push this
Symbol

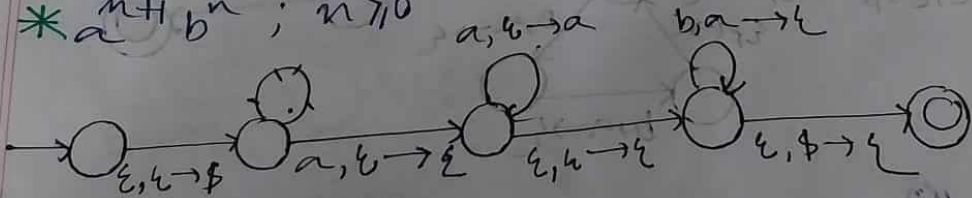
* $a^n b^n; n \geq 0$



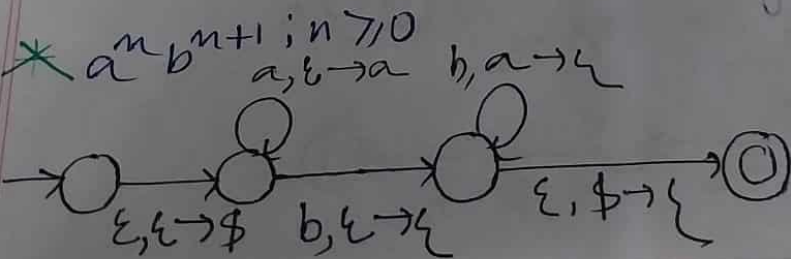
* $a^n b^n$; $n \geq 0$

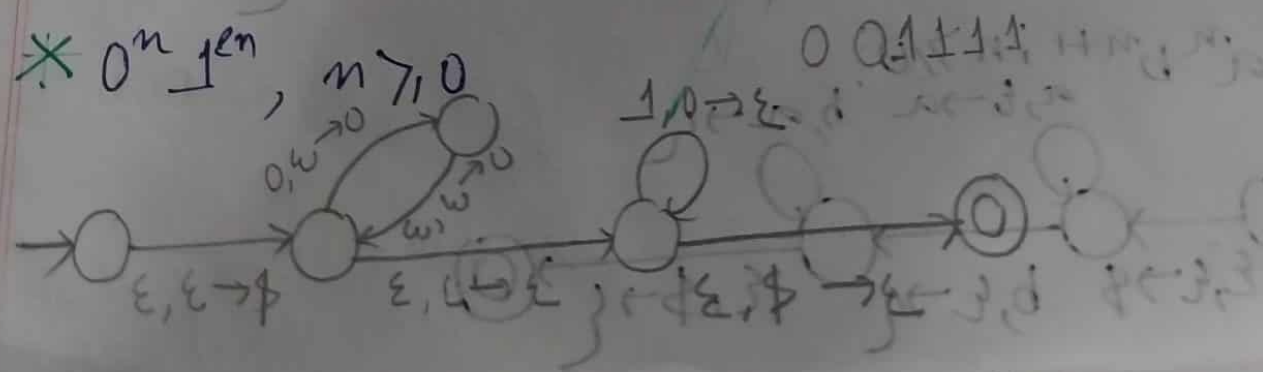
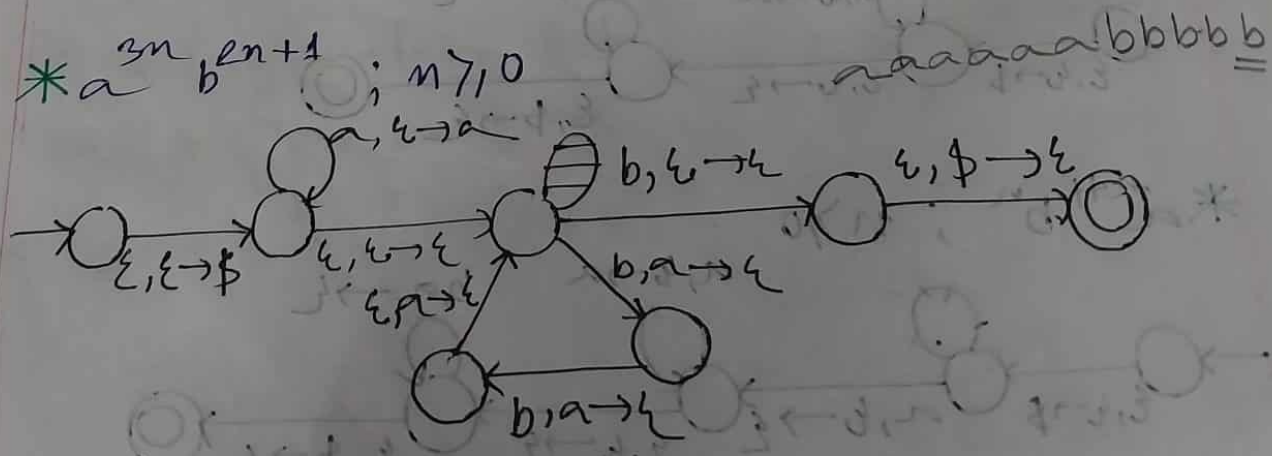
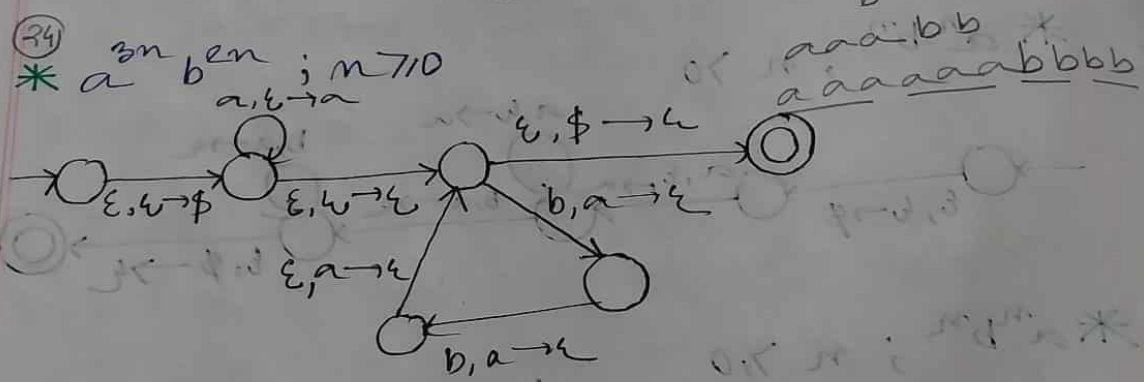
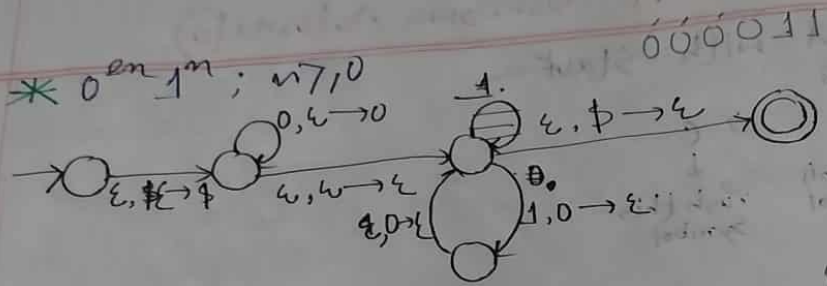


* $a^{n+1} b^n ; n \geq 0$

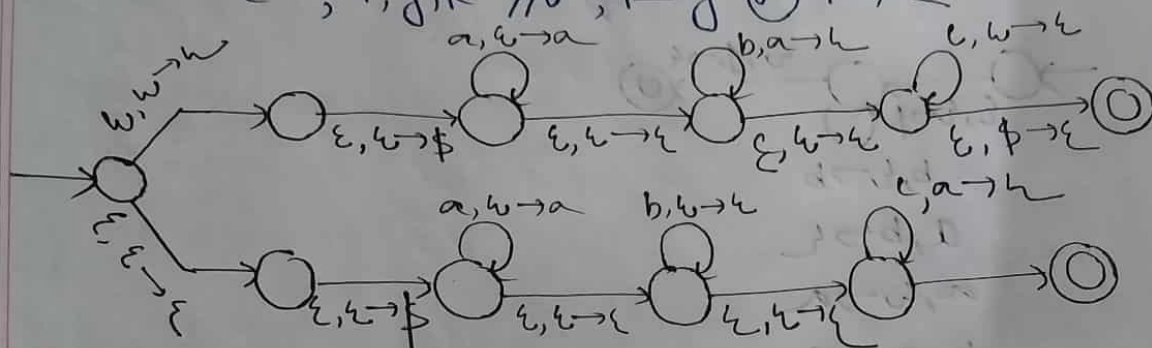


* $a^n b^{n+1}; n \geq 0$
 $a, b \rightarrow a$



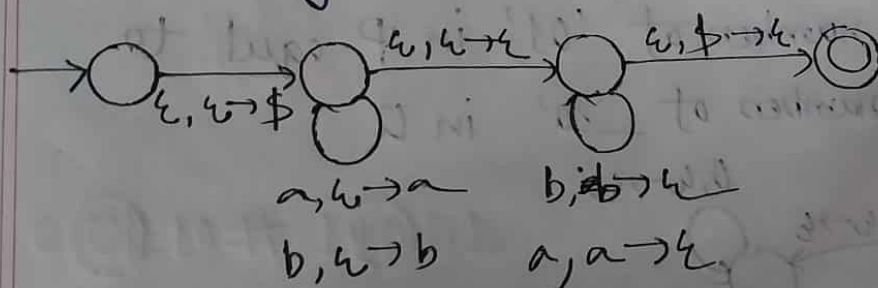


* $a^i b^j c^k$; $i, j, k \geq 0$, $i=j$ (or) $i=k$



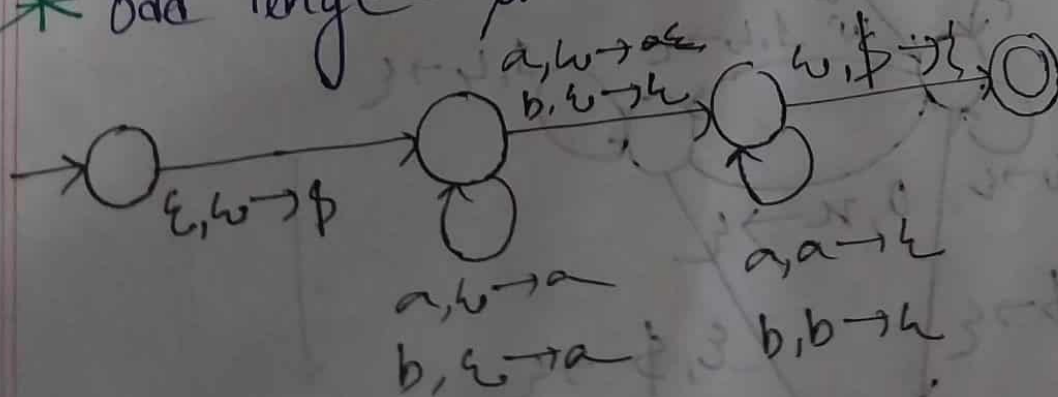
22

* even length palindrome (ww^R) $a^*ba^*b^*a^*$

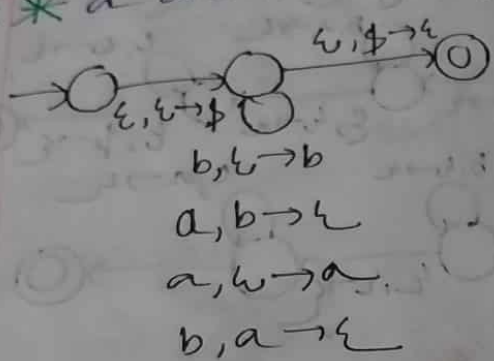


23

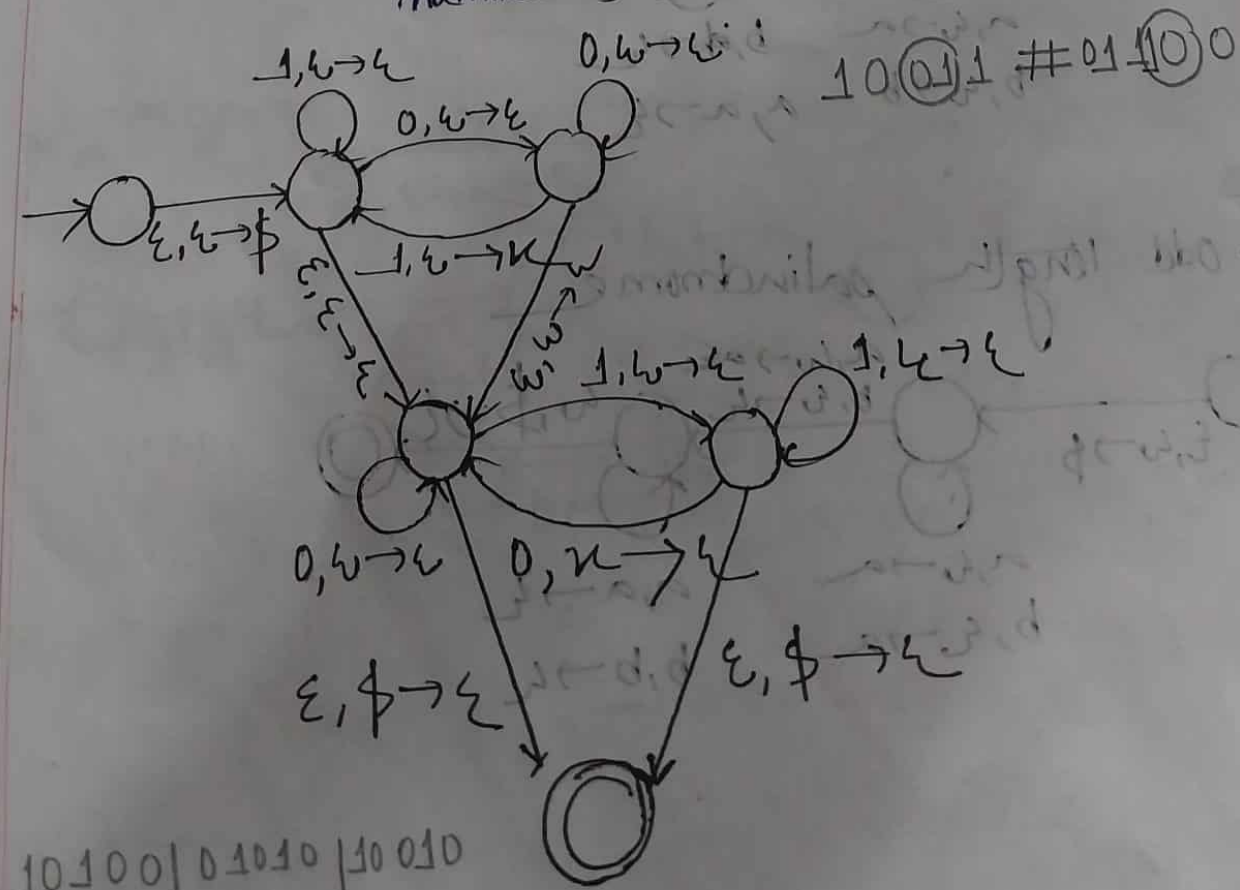
* odd length palindrome $a^*b^*a^*$



* * a and b are equal babba



* * ²³ $p \neq Q$; number of '01' in P equal to number of '10' in Q



10(01)1 # 011(10)0

10100	01010	10010
10101	11010	10110

More Practice Problem

$$* a^m b^n c^n d^m |m, n\rangle = 1$$

Is this true? ... to answer **

$$\checkmark * a^m b^m a^m b^m a^{m+n} |m, n\rangle = 1$$

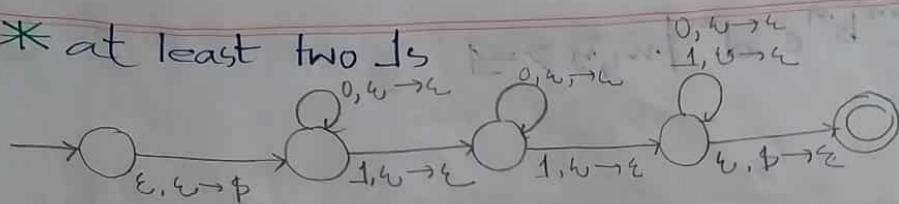
$$\hookrightarrow a^m b^m a^m b^m a^n$$

$$* a^n b^m c^{(n+m)} |n, m\rangle, 1$$

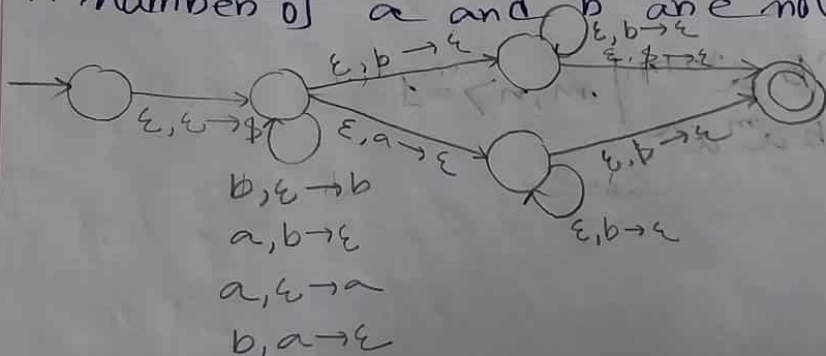
$$* a^{(n+m)} b^m c^n |m, m\rangle, 1$$

$$* a^n b^{(n+m)} c^m |m, m\rangle, 1$$

* at least two 1s



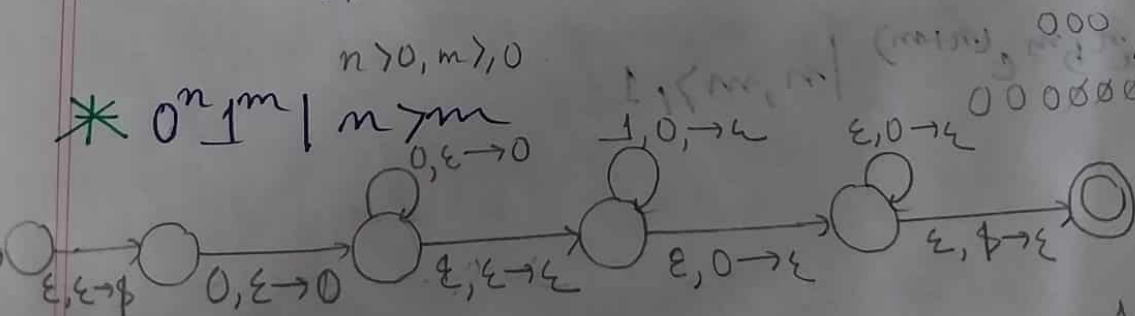
(24) * * number of a and b are not equal



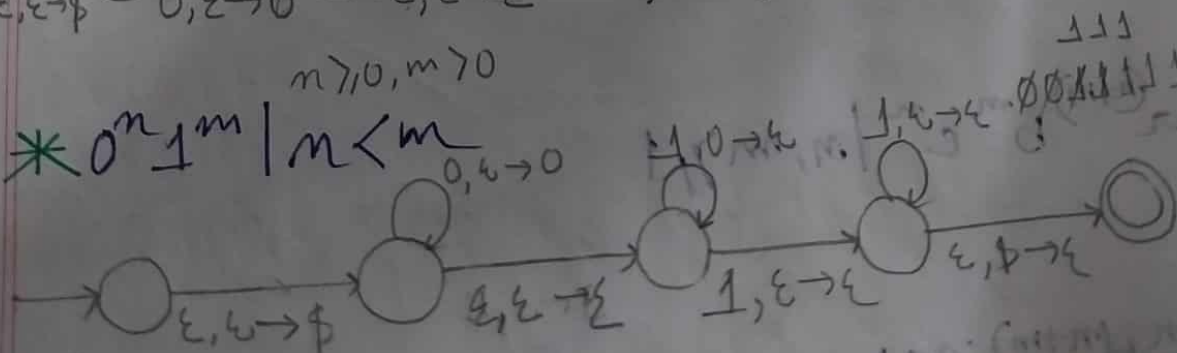
baabbb
baabaa

b	a
b	a
a	a
a	a

* $0^n 1^m \mid n > m$



* $0^n 1^m \mid n < m$



* $0^n 1^m \mid n \geq m$

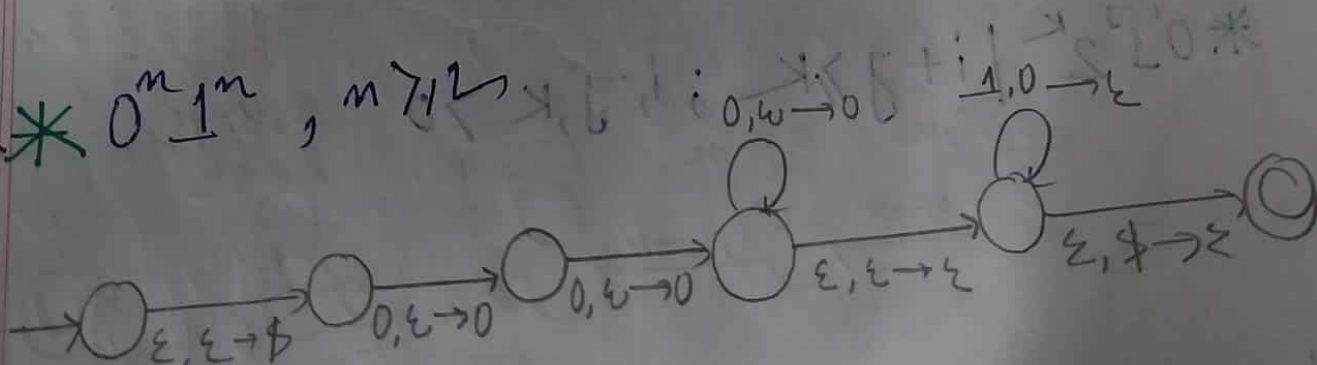
* $0^n 1^m \mid n \geq m$

* {No. n's are greater than no. of y's} *

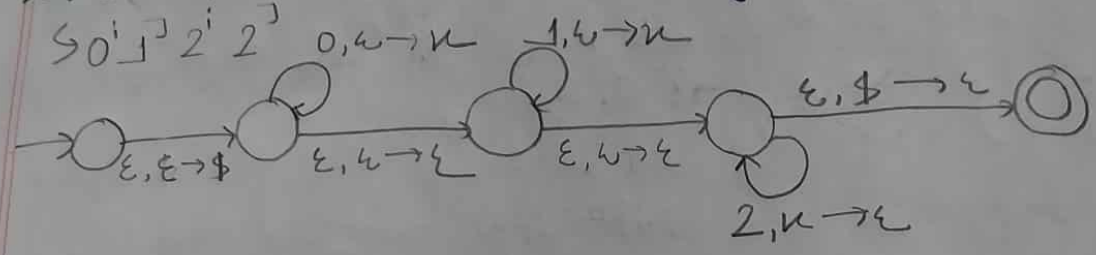
* $x^n, y^m \mid m, n \geq 1 \& m \geq n+2$ *

* Design a PDA to match parenthesis. $((()))$ *

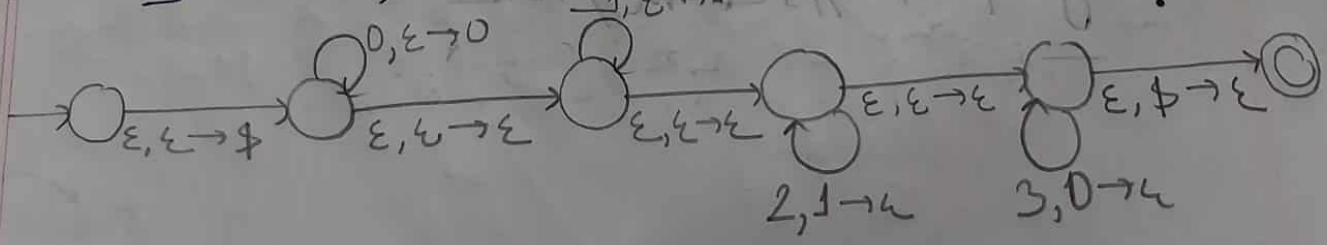
* $0^n 1^n, n \geq 1$ *



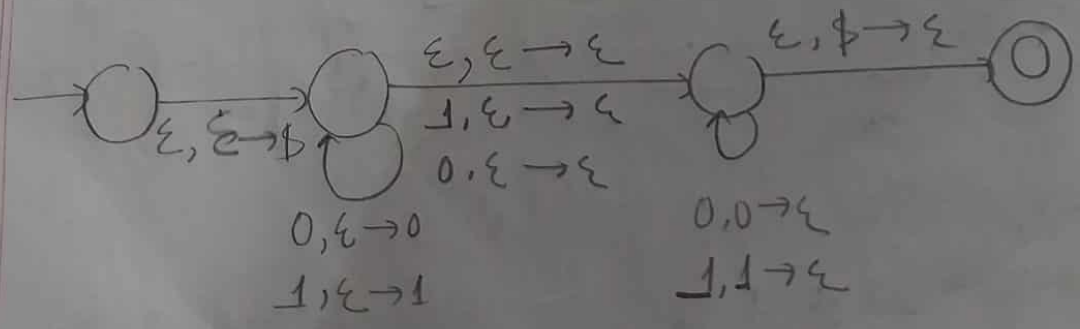
* $0^i 1^j 2^k$; $i, j, k \geq 0$; $i+j=k$ 0 1 1 2 2 2



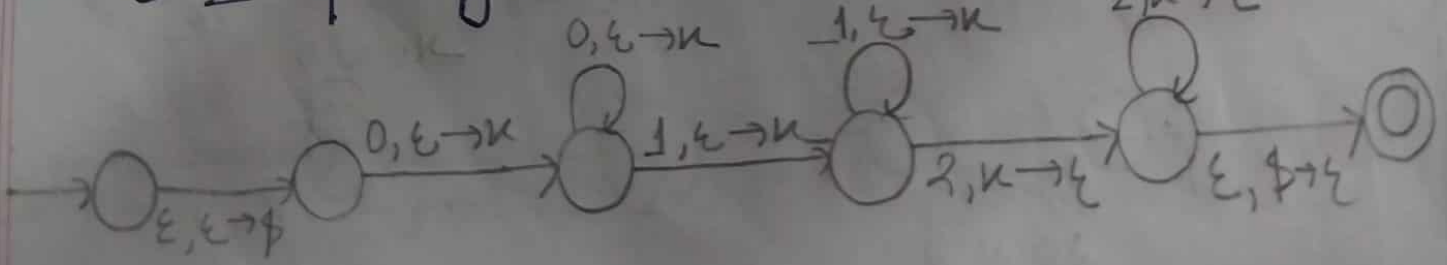
* $0^i 2^j 3^i$; $i, j \geq 0$



* Palindrome (~~www^R~~)

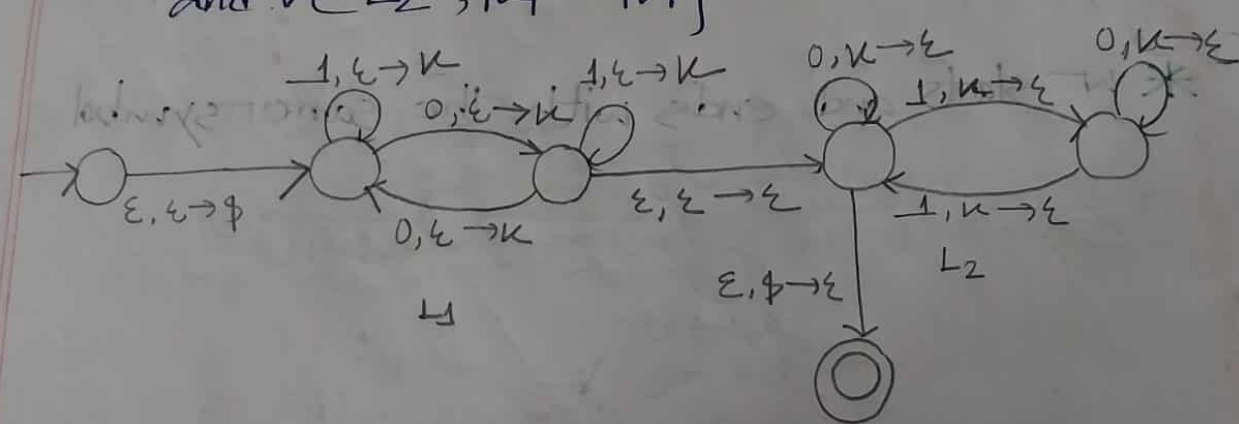


* $0^i 1^j 2^k$ | $i+j \geq k$; $i, j, k \geq 0$

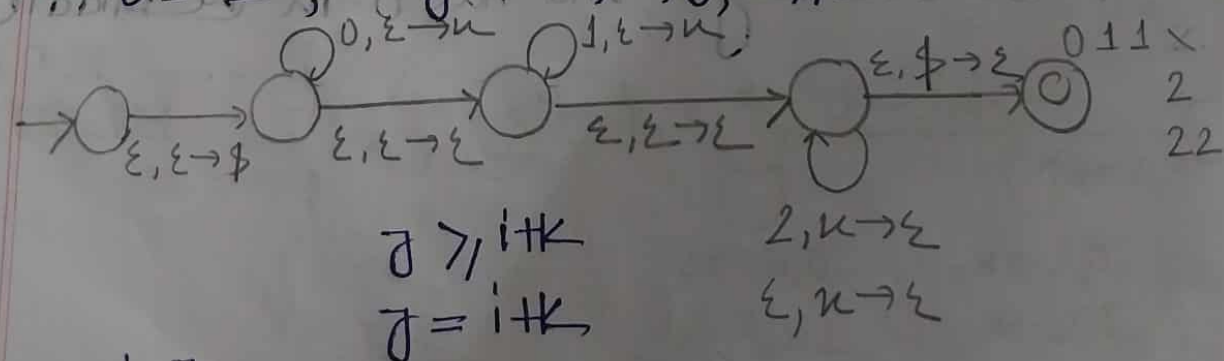


* $L_1 = \{w \in \Sigma^* : w \text{ contains odd numbers of } 0's\}$
 * $L_2 = \{w \in \Sigma^* : w \text{ contains even numbers of } 1's\}$

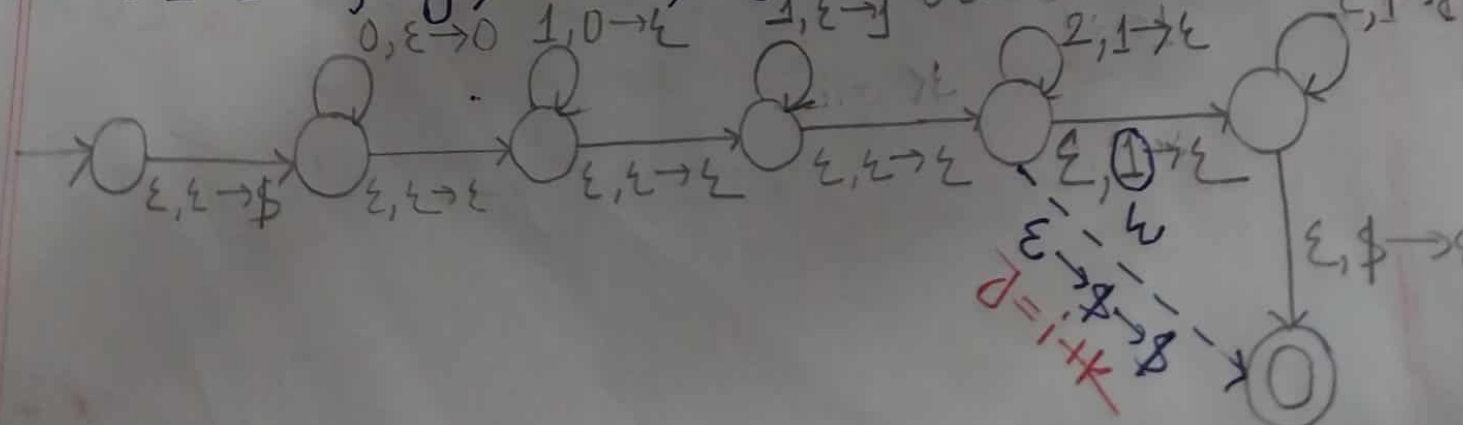
$L = \{w \in \{0,1\}^* : w = uv, \text{ where } u \in L_1 \text{ and } v \in L_2, |u| = |v|\}$



* $0^i 1^j 2^k ; i+j > k, i, j, k \geq 0$

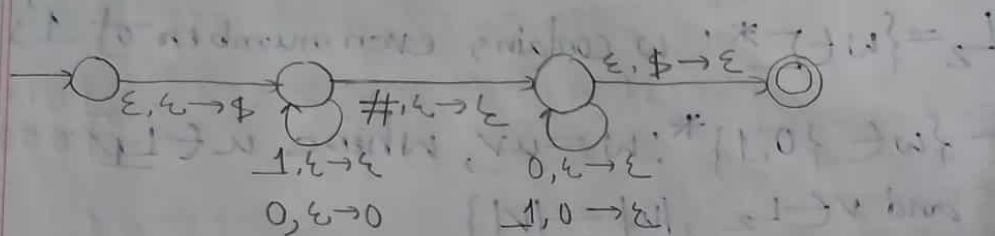


* $0^i 1^j 2^k ; j > i+k, i, j, k \geq 0$



$$10010 = 0101100$$

* $w_1 \# w_2$, where no. of 0's in $w_1 =$ no. of 1's in w_2

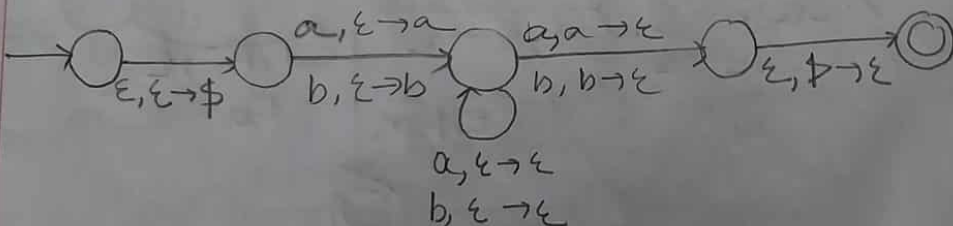


(24)

* w starts and ends with the same symbol

$$a \Sigma^* a$$

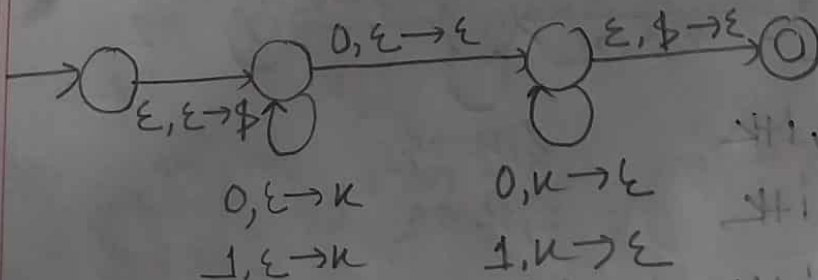
$$b \Sigma^* b$$



* w is of odd length and has 0 in the middle

$$\Sigma 0 \Sigma$$

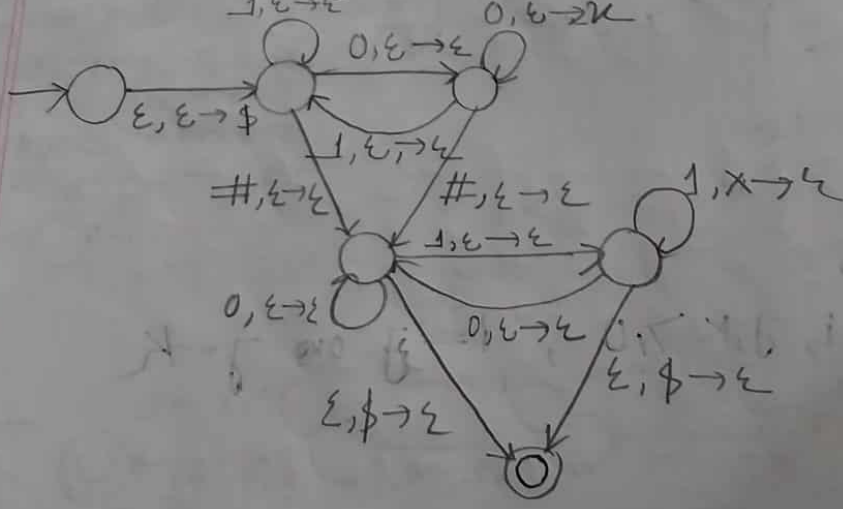
$$\Sigma \Sigma 0 \Sigma \Sigma$$



22 *

*** $L = \{w_1 \# w_2 \mid \text{number of '00' substring in } w_1 \text{ is equal to the number of '11' substring in } w_2\}$

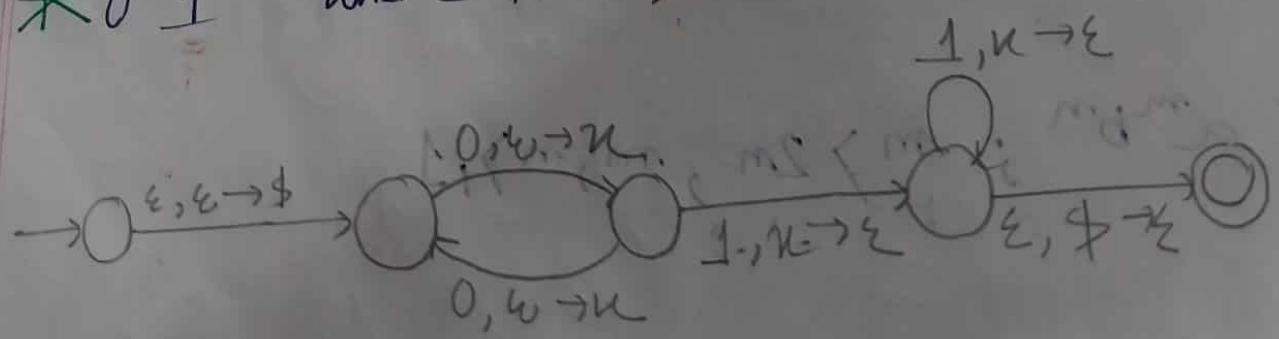
001001101# 00100101110



0000111

22 *

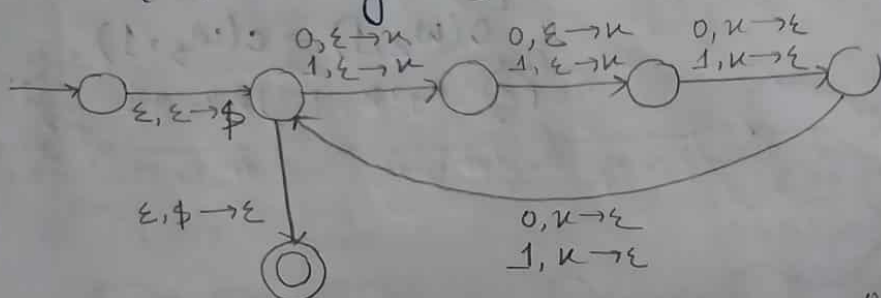
$0^n 1^n$ where n is odd



w,
tring

22

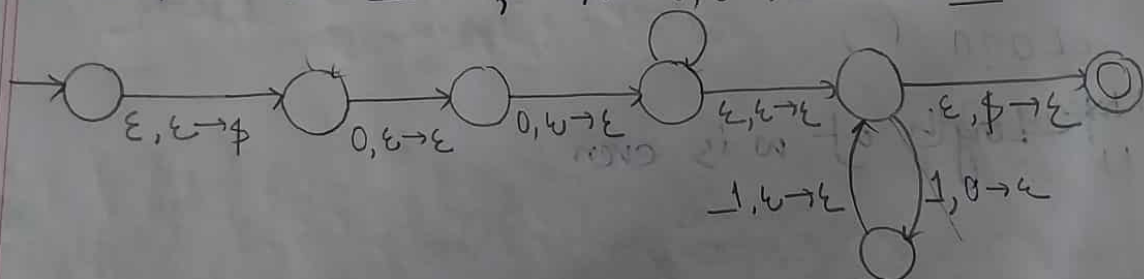
* $L = \{w \mid \text{the length of } w \text{ is divisible by } 4\}$



1001
11000100

22

* $w \mid w = 0^{n+2} 1^n, n \geq 0$



00001111
00011
00

23

* $L = \{w_1 \# w_2 \# w_3 \mid c(w_1, 0) = c(w_2, 0) \text{ or } c(w_2, 1) = c(w_3, 1)\}$

a) 00010#1001#---

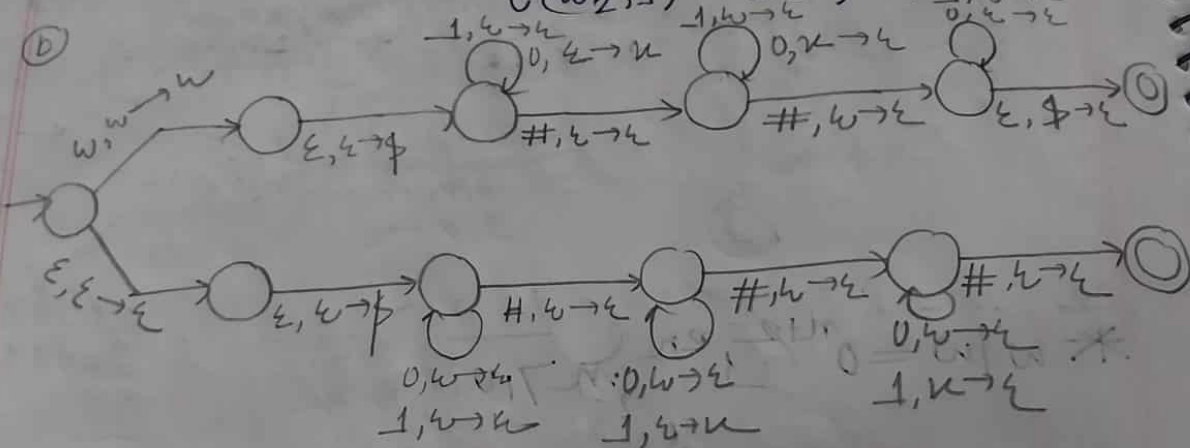
110

101

011

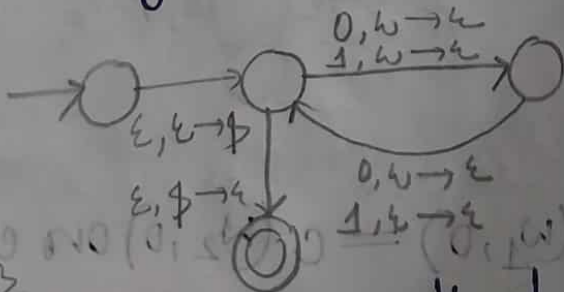
10100#11000111#

23
 * $L = \{w_1 \# w_2 \# w_3 \mid c(w_1, 0) = c(w_2, 0) \text{ or } c(w_2, 1) = c(w_3, 1)\}$



23

* Length of w is even



23

* w contains at least two 1

